

Portfolio project

PHARMA MANUFACTURING DATA PLATFORM

Unified Data Platform for Production, Quality, Process & Supply Risk Analytics

Synthetic pharmaceutical manufacturing data • Azure • Databricks • Snowflake • dbt • Power BI



1. INTRO / PROBLEM STATEMENT / DATASET

From disconnected factory systems to one trusted manufacturing view

Business problem

Production, quality, sensor and supply data lived in separate systems. Teams could not quickly connect batch problems with lab failures or abnormal process conditions. The project creates a governed Gold layer and a Power BI control tower.

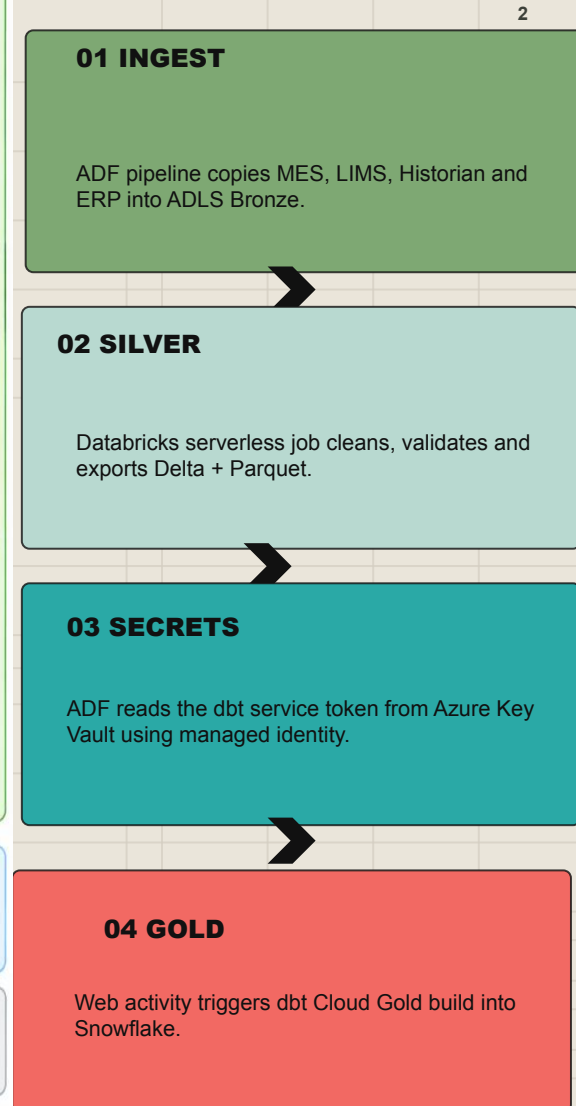
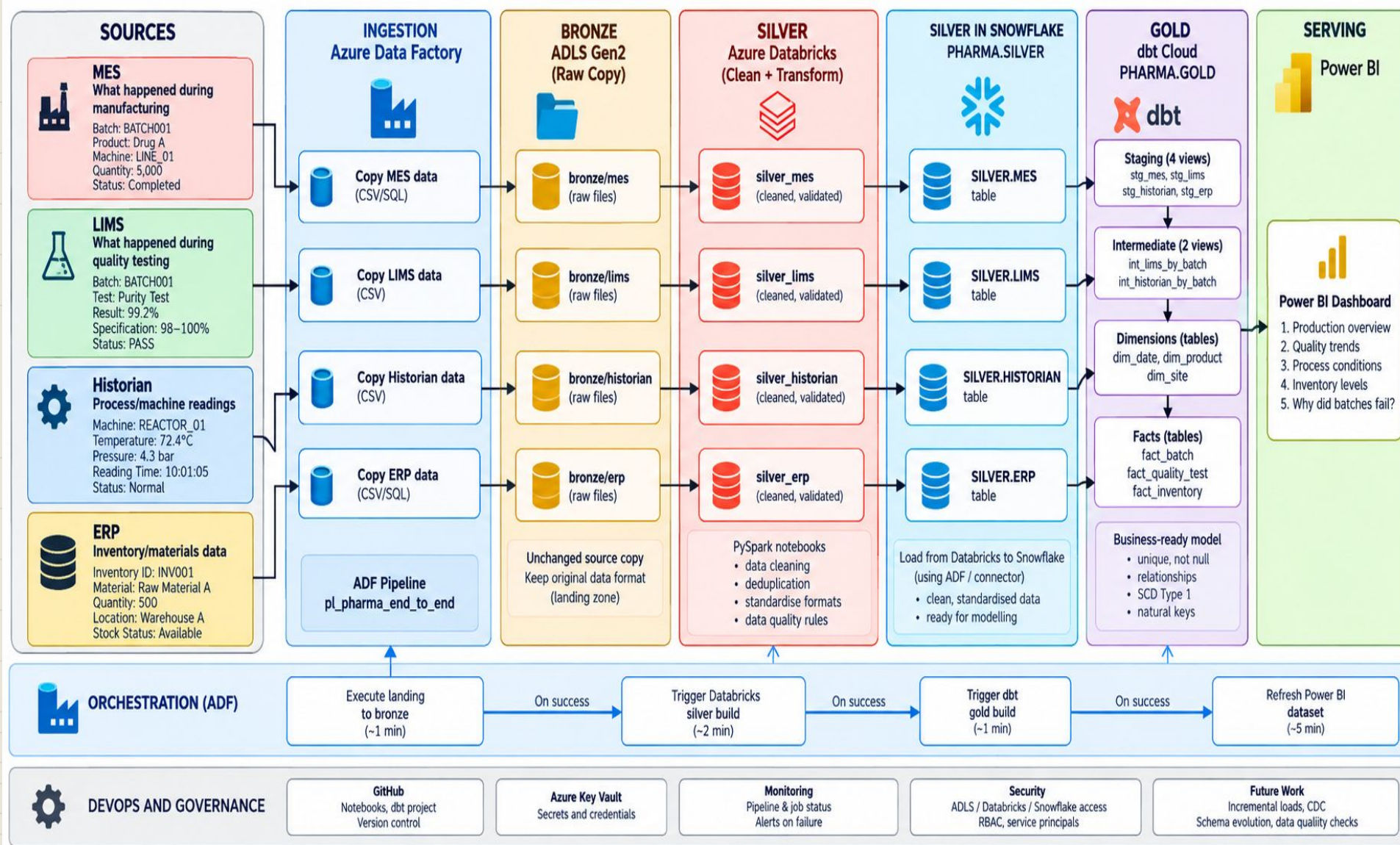
Dataset footprint

MES	LIMS
500 batches	1,000 tests
Historian	ERP
20,133 readings	52 low stock / 10 late

MES manufacturing execution system	What happened during manufacturing Batch: BATCH001 Product: Drug A Machine: LINE_01 Quantity: 5,000 Status: Completed
LIMS laboratory information system	What happened during quality testing Batch: BATCH001 Test: Purity Test Result: 99.2% Specification: 98–100% Status: PASS
HISTORIANS	What happened during the manufacturing process Machine: REACTOR_01 Temperature: 72.4°C Pressure: 4.3 bar Reading Time: 10:01:05 Status: Normal
ERP enterprise resource planning	What happened with inventory/materials Inventory ID: INV001 Material: Raw Material A Quantity: 500 Location: Warehouse A Stock Status: Available

2. INGESTION / ORCHESTRATION / PIPELINE

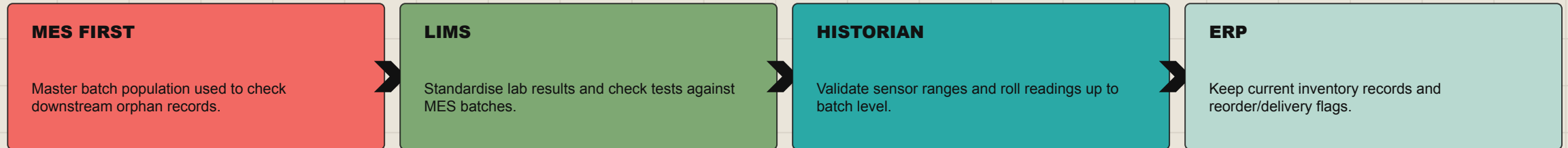
One controlled end-to-end run: bronze → silver → gold



3. ETL / BRONZE TO SILVER

Databricks transforms raw source files into clean, consistent Silver data

Silver build order

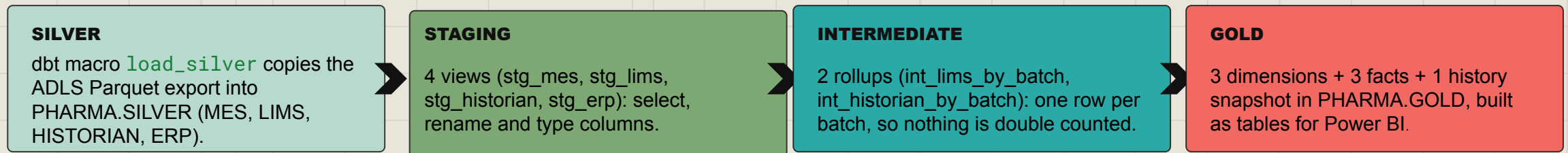


ETL focus

1. Standardise timestamps
2. Cast data types
3. Remove exact duplicates
4. remove leading, ending spaces trim
5. logical flags

4. WAREHOUSE / DATA MARTS

Snowflake + dbt turns Silver data into tested Gold facts and dimensions



Gold star model

Dimensions	Facts
dim_date (731 days)	fact_batch: 1 row per batch (500)
dim_product	fact_quality_test: 1 row per test (1,000)
dim_site	fact_inventory: 1 row per item

Design principle: start from MES and left join supporting data so no valid production batch is dropped.

Testing + governance

Test	What it checks
not_null	Every key has a value
unique	One row per batch, test and dimension key
relationships	Every fact row links to a real dimension
accepted_values	quality_result is only Pass, Fail or OOS

- 8 source tests check the input, and 17 gold tests check the output
- A failed test stops the build, so bad data never reaches Power BI
- Validated: 500 batches, 1,000 tests and 20,133 readings, with none lost or double counted

5.Data Modelling

mes

BATCH_ID	PRODUCT_CODE	SITE	PLANNED_QTY	ACTUAL_QTY	STATUS	DEVIATION_FACTOR	START_TIME	END_TIME
BATCH-00123	PH_NEU_01	Dublin	10,000	9,650	Completed	No	2025-03-14 06:00	2025-03-14 18:30

PRODUCT_CODE	PRODUCT_NAME	THERAPY_AREA	DATE_DAY	YEAR	MONTH	DAY_NAME
PH_NEU_01	Neurocalm 50 mg tablets	Neurology	2025-03-14	2025	March	Friday

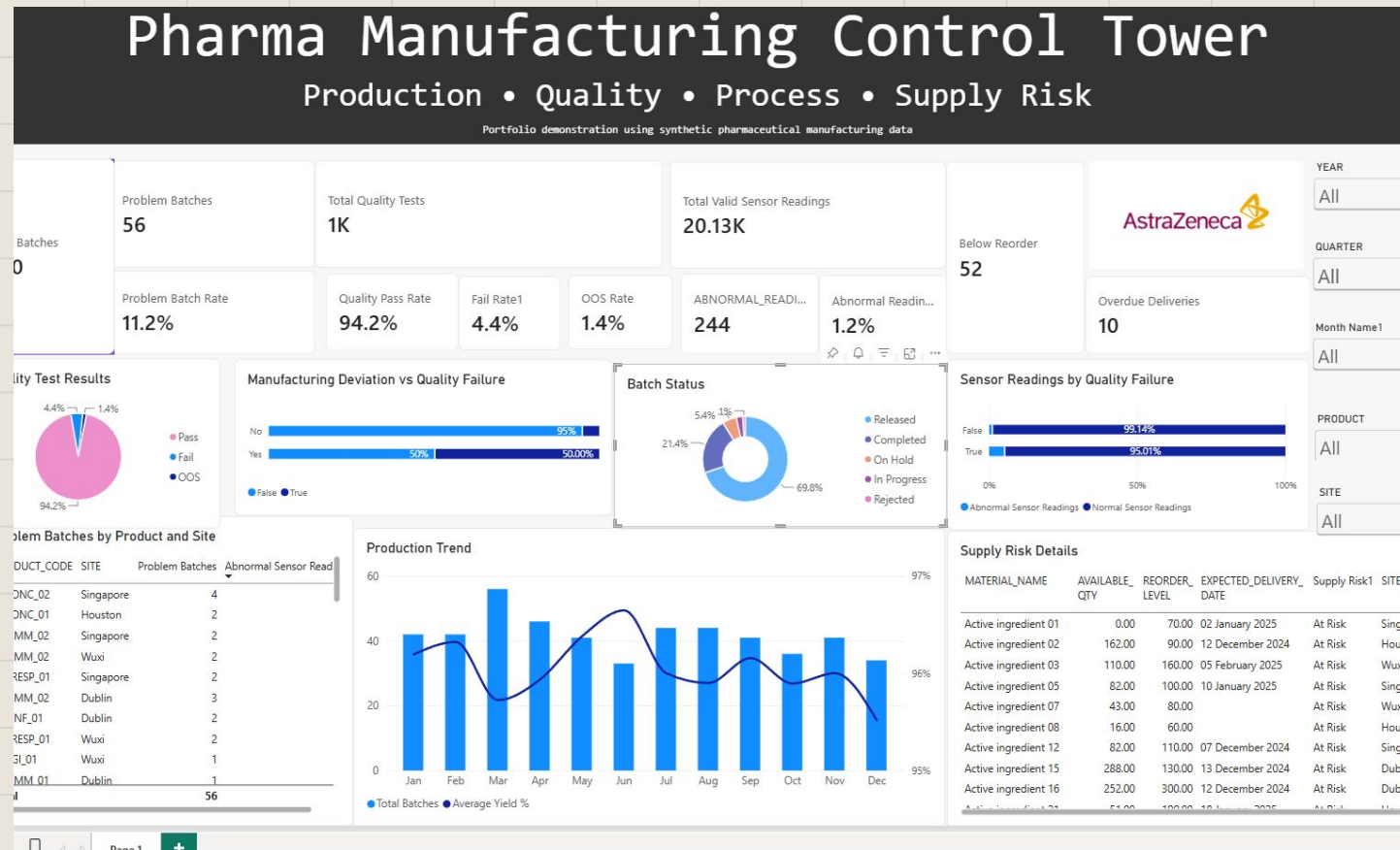
INVENTORY_ID	MATERIAL	SITE	AVAILABLE_QTY	REORDER_LEVEL	BELOW_REORDER	DELIVERY_OVERDUE
INV-0042	API-NEU (active ingredient for PH_NEU_01)	Dublin	120 kg	200 kg	Yes	No

dim_product shows that PH_NEU_01 is Neurocalm 50 mg, **dim_site** shows it was manufactured at the Dublin site, **dim_date** shows production started on 14 March 2025, **fact_batch** shows batch BATCH-00123 produced 9,650 units from 10,000 planned with 96.5% yield, 40 process readings and 2 abnormal readings, **fact_quality_test** shows its Assay passed but Dissolution failed with a result of 78 against a minimum of 80, and **fact_inventory** shows INV-0042 (API-NEU) had 120 kg available against a 200 kg reorder level, so it was below reorder level but delivery was not overdue.”

INV-0042, which is **API-NEU**, the active ingredient for product **PH_NEU_01**, is held at the Dublin site with 120 kg available against a reorder level of 200 kg, meaning it is below the reorder level, while its delivery is not overdue.

5. DASHBOARD / BUSINESS VALUE

The dashboard turns disconnected manufacturing data into early warning signals — showing not just *what failed*, but *where to investigate and why*.



Key dashboard findings

Problem batches

56 / 11.2%

Quality pass rate

94.2%

Abnormal sensor rate

1.2%

Supply risk

52 low / 10 late

Key Insights from the Control Tower

Deviations are a strong warning signal
Batches with deviations had a **50% quality-failure rate**, compared with only **~5%** for normal batches.

Sensor behaviour reveals quality risk
Failed batches showed **5x more abnormal sensor readings** — **5% vs 0.9%** — linking process conditions with quality outcomes.

6. CHALLENGES

The important problems we found and how we handled them

Challenge	How we solved it
18 lab status mismatches	Trusted the measured value against the limits, kept the lab status, and flagged every mismatch
15 lab orphans + 40 sensor orphans	Flagged in silver, excluded in gold with inner joins to MES batches
Joining 20,133 readings to 500 batches would multiply rows	Rolled up tests and readings to one row per batch before joining
Serverless Databricks blocked storage keys	Unity Catalog with a managed identity (Access Connector): no keys anywhere
dbt service user locked after failed password logins	Key-pair authentication for DBT_SERVICE
Secrets across 5 services on shared accounts	Tokens in Azure Key Vault, read by ADF's managed identity; roles assigned by Object ID
ADF step started too early	Connected every activity with on-success dependencies

Honest limitation — The ADF Web activity starts the dbt job but does not yet wait for the dbt run to finish. A future loop can check the dbt API until completion.

7. FUTURE WORK

Move from daily batch reporting to stronger production-grade operations

Incremental at scale

Apply our working CDC demo to all four sources: ADF watermarks, Auto Loader + Delta MERGE, COPY INTO without reload, dbt incremental models.

Monitoring and alerts

ADF waits for the dbt run result, and sends Teams or email alerts on any failure. dbt source freshness checks catch a missed load.

Security

Snowflake storage integration instead of a SAS token; service principal instead of a personal Databricks token; branch protection on main.

Data quality and schema evolution

Business-rule tests (Great Expectations or dbt_expectations): sensor ranges, date order. Detect new or renamed columns(schema changes)

Rejected data handling

Quarantine tables for failed rows, with a daily review by the quality team, and a clear rule to fix, reprocess or accept each record.

Reporting and real time

Power BI row-level security by site, and alerts on problem batches. Stream Historian sensor data through Event Hubs.

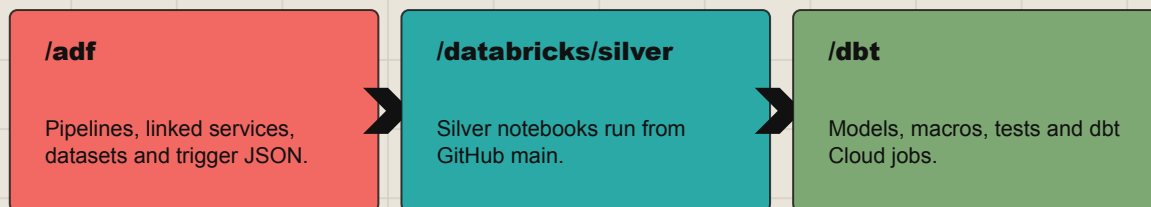
Future direction

The platform is already full-refresh; the next evolution is incremental, monitored and more secure production delivery.

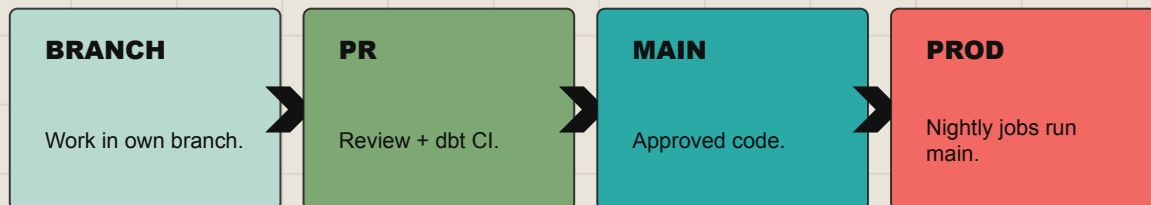
8. CI / CD

One repository, pull requests, automated dbt checks and approved main branch deployment

Repository structure



Branching rule



CI/CD design

Layer	CI before merge	CD from main
ADF	Debug + PR review	Publish ARM templates
Databricks	Run notebooks + PR	silver_build job uses main
dbt	PR job builds modified models	Gold build from main

Security

Secrets stored in Azure Key Vault, not in ADF JSON or GitHub.
 ADF managed identity has read-only secret access.
 Known limitation: private GitHub free plan cannot enforce branch protection; team rule used instead.